Task Submission Document - Iris Classification

Task No: _1___ | Student Name: Jeswanth Reddy Reddem | Date: _____20/05/2026_____________

Important: Before final submission, replace all placeholder links below with your actual GitHub, Google Drive, and notebook links. Ensure Google Drive sharing is set to "Anyone with the link can view".

1-Page Executive Summary

Project Goal: Build a supervised machine learning classification model to predict iris flower species using sepal length, sepal width, petal length, and petal width.

Dataset: Classic Iris dataset containing three species: Iris-setosa, Iris-versicolor, and Iris-virginica. The dataset was split into training and testing sets using stratification to preserve class distribution.

EDA and Visualization: Exploratory data analysis was performed using class distribution charts and feature scatter plots. Petal length and petal width showed strong class separability, especially for Iris-setosa.

Models Trained: k-Nearest Neighbors, Logistic Regression, and Decision Tree classifiers were trained and compared.

Evaluation Metrics: Models were evaluated using accuracy, precision, recall, F1-score, and confusion matrix. The best model was selected based on test accuracy and macro/weighted performance.

Best Model: Logistic Regression was selected as the best model in this run, achieving 93.33% test accuracy. The trained model was saved using both joblib and pickle formats.

Deliverables: The submission contains a well-structured notebook, source code, saved model files, generated plots/screenshots, README, requirements file, and inference example.

Submission Links

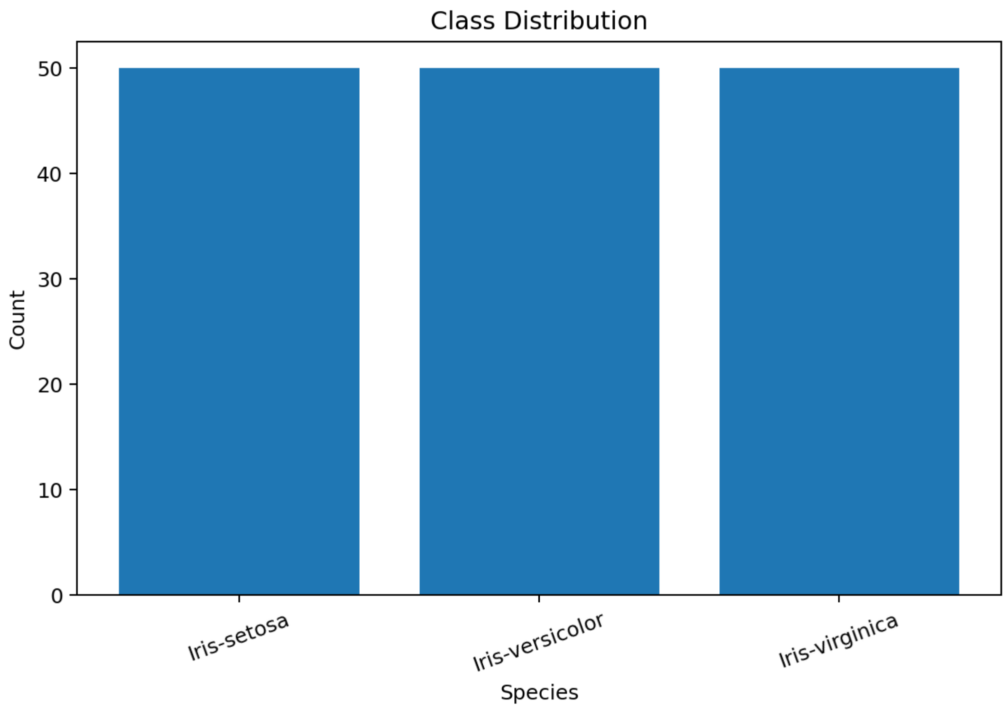
Artifact	Required Link	Status
GitHub Repository	GitHub repo link	Added
Notebook Link	GitHub notebook link	Added
Saved Model File	GitHub model file link	Added
Screenshots/Plots Folder	GitHub screenshots folder link	Added

Artifacts Included

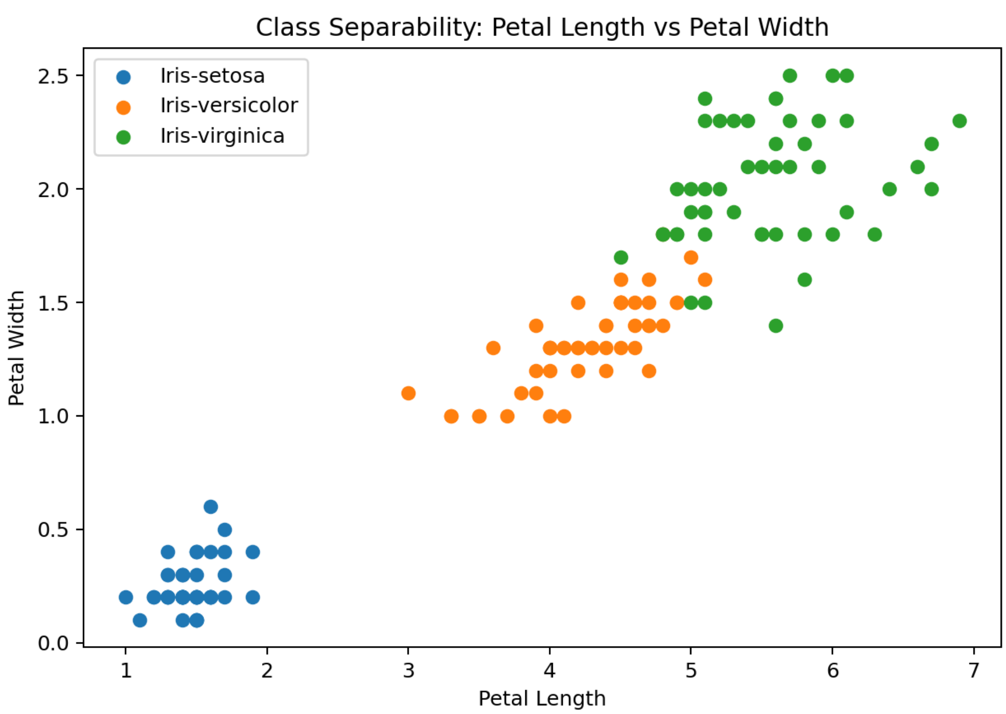
- Notebook: notebooks/Iris_Classification.ipynb
- Training script: train_model.py
- Inference script: inference.py
- Saved models: models/best_iris_model.joblib and models/best_iris_model.pkl
- Plots/screenshots: plots/class_distribution.png, plots/petal_scatter.png, plots/sepal_scatter.png, plots/metrics_comparison.png, plots/confusion_matrix_best_model.png
- README: README.md
- Requirements: requirements.txt
- Metrics summary: metrics_comparison.csv

Screenshots / Plots

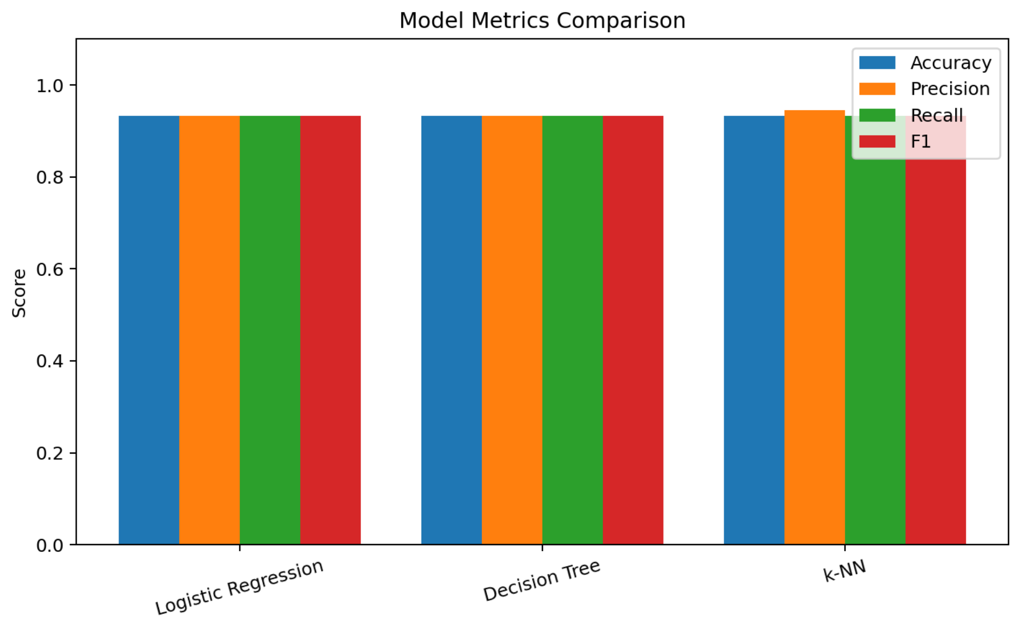
Class Distribution



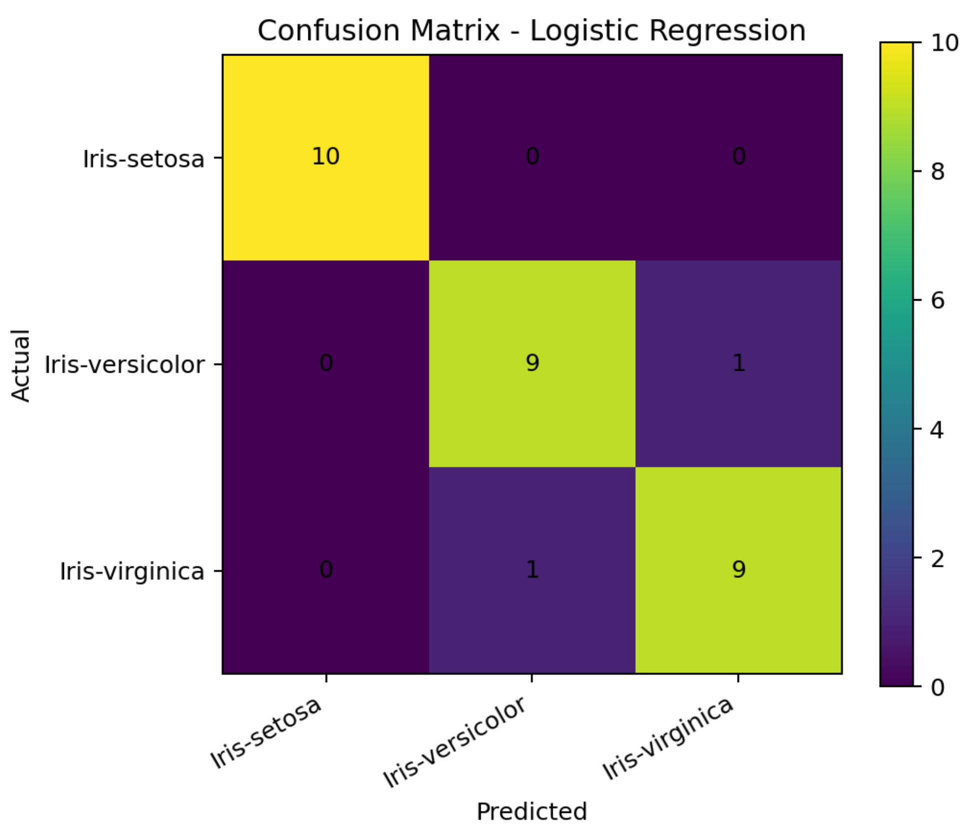
Petal Feature Separability



Metrics Comparison



Confusion Matrix - Best Model



Reproducibility Commands

```
# 1. Install dependencies
pip install -r requirements.txt

# 2. Train and evaluate models
```

```
python train_model.py

# 3. Run example inference
python inference.py

# 4. Open the notebook
jupyter notebook notebooks/Iris_Classification.ipynb
```

Package Versions

- Python 3.x
- pandas
- numpy
- matplotlib
- seaborn
- scikit-learn
- joblib
- jupyter

Final Submission Checklist

☐ Push notebooks and code to GitHub.

☐ Upload large files or final ZIP to Google Drive if required.

☐ Set Google Drive sharing permission to anyone with the link can view.

☐ Paste links clearly on the Task Submission page with task number.

☐ Attach or upload this DOC/PDF submission document as required.

☐ Verify that README includes package versions and exact commands.